Classification Knee Osteoporosis X-Ray

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March 31, 2026

Problem

Osteoporosis is a chronic bone disease characterized by decreased bone density and deterioration of bone microstructure, leading to increased fragility and fracture risk. It is considered a major public health concern, especially among elderly populations. Early detection of osteoporosis is essential because appropriate intervention can significantly reduce the risk of fractures and improve quality of life.

In clinical practice, osteoporosis is typically diagnosed using DEXA Scan, which provides quantitative measurements of bone mineral density. However, DEXA scans are not always available in all healthcare settings due to cost, equipment requirements, and accessibility limitations. In contrast, X-ray imaging is widely available and commonly used as a first-line imaging tool.

Despite its availability, interpreting osteoporosis from X-ray images is not straightforward. The differences between normal and osteoporotic bone often appear as subtle variations in texture, brightness, and structural patterns. These differences may not be clearly visible, especially in early or mild cases. As a result, interpretation can be subjective and dependent on the experience of the clinician.

The problem addressed in this project is the development of an automated system capable of classifying knee X-ray images into two categories: normal and osteoporosis. The goal is to investigate whether a deep learning model can learn discriminative features from X-ray images and provide reliable classification results.

This project focuses on knee X-ray images because the knee joint is commonly imaged and contains important bone structures such as the distal femur and proximal tibia, where bone density changes can be observed. The system is designed as a prototype for educational and research purposes, demonstrating how computer vision techniques can assist in medical image analysis.

Data

The dataset used in this project was obtained from a publicly available source on Kaggle titled *Multi-Class Knee Osteoporosis X-Ray Dataset*. The dataset consists of knee X-ray images categorized into multiple bone density conditions. For the purpose of this project, only two classes were selected:

- Normal
- Osteoporosis

The osteopenia class, which represents an intermediate stage of bone density loss, was excluded to simplify the classification task and increase the separability between classes.

The final dataset used in this study contains a total of **1,573 images**, including:

- 780 images labeled as normal
- 793 images labeled as osteoporosis

The dataset is relatively balanced, which is beneficial for training a classification model because it reduces bias toward one class.

To ensure proper evaluation, the dataset was divided into three subsets:

- Training set (70%)
- Validation set (20%)
- Test set (10%)

The distribution is as follows:

Normal class:

- Training: 546 images
- Validation: 156 images
- Test: 78 images

Osteoporosis class:

- Training: 555 images
- Validation: 158 images
- Test: 80 images

All images were organized into a folder-based structure compatible with deep learning frameworks, where each class is stored in separate directories under train, validation, and test folders.

Before training, images were resized to a consistent resolution of 224 × 224 pixels to match the input requirements of the model. The dataset may also include variations in brightness, contrast, and image quality, which reflect real-world conditions and contribute to the robustness of the model.

The use of a publicly available dataset ensures compliance with ethical guidelines and project requirements, as the dataset does not contain identifiable personal information and is intended for academic use.

Methodology

Pipeline Overview

The proposed system is developed as an end-to-end image classification pipeline for detecting bone density conditions from knee X-ray images. The overall workflow consists of the following stages:

Input → Preprocessing → Model Training → Prediction → Evaluation → Failure Analysis

This pipeline enables automated classification of bone conditions into normal and osteoporotic categories.

Input Data

The input of this system consists of knee X-ray images collected from a public dataset. The dataset includes two classes:

- **Normal**: Images representing healthy bone structure
- **Osteoporosis**: Images showing reduced bone density

A total of 1,573 images were used in this study, with a balanced distribution between the two classes.

Data Preprocessing

Before training, all images were preprocessed to ensure consistency and improve model performance. The preprocessing steps include:

- Resizing images to **224 × 224 pixels**
- Normalizing pixel values
- Splitting the dataset into:
    - Training set (70%)
    - Validation set (20%)
    - Test set (10%)

Additionally, data augmentation techniques such as flipping and brightness adjustment were applied to enhance model generalization.

Model Training

The classification model is based on a pretrained deep learning architecture, specifically **YOLO11 for image classification**. Transfer learning was applied to leverage pretrained weights and improve learning efficiency.

The model was trained using the following configuration:

- Model: YOLO11n-cls
- Epochs: 70
- Image size: 224 × 224
- Batch size: 16

The model learns discriminative features from bone density patterns in X-ray images to distinguish between normal and osteoporotic conditions.

Prediction

After training, the model was used to predict unseen test images. For each input image, the model outputs:

- Predicted class (normal or osteoporosis)
- Confidence score (probability of prediction)

Example output:

Prediction: NORMAL

Confidence: 59.64%

System Summary

The developed system successfully demonstrates an automated approach for bone density classification using deep learning. The pipeline integrates preprocessing, model training, prediction, and evaluation into a complete workflow, making it suitable for medical image analysis tasks at a prototype level.

Evaluation

Accuracy was used as the primary metric to evaluate the overall performance of the model, measuring the proportion of correctly classified images. A higher accuracy indicates better classification capability. In addition, a confusion matrix was employed to provide a detailed visualization of classification results, showing how many images were correctly or incorrectly predicted for each class. To further understand model behavior, confidence scores were analyzed, as they indicate how certain the model is about its predictions. Low-confidence predictions suggest uncertainty and are often associated with difficult or ambiguous cases. Furthermore, qualitative evaluation was conducted by visually inspecting individual predictions, allowing a deeper understanding of how the model performs on different types of images beyond numerical metrics alone.

Failure Analysis

Case 1: Low-confidence correct prediction

The model correctly predicted the image as normal but with relatively low confidence (59.64%), indicating uncertainty in distinguishing between classes. This occurs because the visual differences between normal and osteoporosis are subtle, and the image may contain features that are similar to both categories. This issue can be improved by increasing dataset diversity and including more representative samples to help the model learn more robust features.

Case 2: Low Contrast Images

The model may struggle to classify images with low contrast because bone structures and density patterns are less visible. Important features such as texture and brightness differences become unclear, making it difficult for the model to extract meaningful information. This can be improved by applying contrast enhancement techniques or preprocessing methods to improve feature visibility.

Case 3: Borderline Cases

The model may misclassify borderline cases where mild osteoporosis appears visually similar to normal bone. In such cases, the difference between classes is minimal, leading to confusion during prediction. This can be improved by including more borderline examples in the training dataset or introducing an additional intermediate class to help the model learn finer distinctions.

Case 4: Image Quality Issues

The model performance may decrease when images are blurred or contain noise, as these factors reduce the clarity of bone structures. This makes it difficult for the model to detect patterns related to bone density. This can be improved by filtering out low-quality images or applying image enhancement techniques.

Case 5: Background Influence

The model may focus on irrelevant regions outside the bone area because it processes the entire image. This can negatively affect prediction accuracy, as important features may be overlooked. This can be improved by cropping the region of interest (ROI) around the knee or applying segmentation techniques to guide the model's focus.

Limitations and Ethics

Limitations

- The model is based only on X-ray images and cannot replace clinical diagnosis
- Subtle differences between classes are difficult to detect
- Dataset limitations may affect generalization
- The model lacks interpretability

Ethics

- The dataset is publicly available and anonymized
- No personal data is used
- The system is intended for research and educational purposes only
- It should not be used for real medical diagnosis

References

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